C

CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 7.01

Differentiation & Rates of Change

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 D	Q3 D	Q4 D	Q5 A
Q6 D	Q7 D	Q8 A	Q9 D	Q10 D
Q11 C	Q12 C	Q13 B	Q14 A	Q15 A
Q16 B	Q17 A	Q18 A	Q19 C	Q20 D
Q21 A	Q22 D	Q23 A	Q24 A	Q25 D

Differentiation & Rates of Change Practice Exercises

1 MCQ

Find the equation of the tangent line to the curve $y = (2x+1)^3$ at the point where $x = 0$.

A $y = 2x + 1$

B $y = 6x - 1$

C $y = 6x + 1$ ✓

D $y = 6x + 3$

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(2x+1)^2 \cdot (2) = 6(2x+1)^2$. At $x = 0$, slope $m = 6$. Equation: $y - 1 = 6(x - 0) \implies y = 6x + 1$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: $y = 6x + 1$. **[A1]**

2 MCQ

Find the equation of the tangent line to the curve $y = (3x+1)^3$ at the point where $x = 0$.

A $y = 9x + 3$

B $y = 9x - 1$

C $y = 3x + 1$

D $y = 9x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(3x+1)^2 \cdot (3) = 9(3x+1)^2$. At $x = 0$, slope $m = 9$. Equation: $y - 1 = 9(x - 0) \implies y = 9x + 1$. [M1][A1]

3 MCQ

Find the equation of the tangent line to the curve $y = (4x+1)^3$ at the point where $x = 0$.

A $y = 12x + 3$

B $y = 12x - 1$

C $y = 4x + 1$

D $y = 12x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(4x+1)^2 \cdot (4) = 12(4x+1)^2$. At $x = 0$, slope $m = 12$. Equation: $y - 1 = 12(x - 0) \implies y = 12x + 1$. [M1][A1]

4 MCQ

Find the equation of the tangent line to the curve $y = (5x+1)^3$ at the point where $x = 0$.

A $y = 15x + 3$

B $y = 15x - 1$

C $y = 5x + 1$

D $y = 15x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(5x+1)^2 \cdot (5) = 15(5x+1)^2$. At $x = 0$, slope $m = 15$. Equation: $y - 1 = 15(x - 0) \implies y = 15x + 1$. [M1][A1]

5 MCQ

Find the equation of the tangent line to the curve $y = (6x+1)^3$ at the point where $x = 0$.

A $y = 18x + 1$ ✓

B $y = 6x + 1$

C $y = 18x - 1$

D $y = 18x + 3$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(6x+1)^2 \cdot (6) = 18(6x+1)^2$. At $x = 0$, slope $m = 18$. Equation: $y - 1 = 18(x - 0) \implies y = 18x + 1$. [M1][A1]

6 MCQ

Find the equation of the tangent line to the curve $y = (7x+1)^3$ at the point where $x = 0$.

A $y = 21x - 1$

B $y = 7x + 1$

C $y = 21x + 3$

D $y = 21x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(7x+1)^2 \cdot (7) = 21(7x+1)^2$. At $x = 0$, slope $m = 21$. Equation: $y - 1 = 21(x - 0) \implies y = 21x + 1$. [M1][A1]

7 MCQ

Find the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.

A $y = 8x + 1$

B $y = 24x + 3$

C $y = 24x - 1$

D $y = 24x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(8x + 1)^2 \cdot (8) = 24(8x + 1)^2$. At $x = 0$, slope $m = 24$. Equation: $y - 1 = 24(x - 0) \implies y = 24x + 1$. [M1][A1]

8 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 3$ cm?

A 18π cm^3/s ✓

B 36π cm^3/s

C 9π cm^3/s

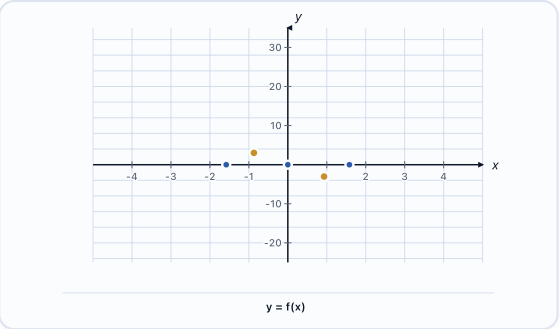
D $\frac{18}{2}\pi$ cm^3/s

WORKED SOLUTION & VERIFICATION

$V = \frac{4}{3}\pi r^3 \implies \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 3$, $\frac{dV}{dt} = 4\pi(3)^2(0.5) = 2\pi(9) = 18\pi$ cm^3/s. [M1][A1]

9 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



A 19 units per second

B $\frac{19}{3}$ units per second

C 21 units per second

D 57 units per second ✓

WORKED SOLUTION & VERIFICATION

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

1. Differentiate y with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$$

2. Evaluate $\frac{dy}{dx}$ at $x = 2$:

$$\left.\frac{dy}{dx}\right|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$$

3. Multiply by $\frac{dx}{dt} = 3$:

$$\frac{dy}{dt} = 19 \times 3 = 57 \text{ units per second}$$

Common errors:

- 19 corresponds to finding $\frac{dy}{dx}$ only and neglecting $\frac{dx}{dt}$.
- $\frac{19}{3}$ corresponds to inverting the chain rule (dividing instead of multiplying by $\frac{dx}{dt}$).
- 21 results from incorrectly differentiating $2x^3$ as $6x$ instead of $6x^2$. [M1][A1]

10 MCQ

Let $f(x) = \dfrac{140x^2 + 10x - 10}{x^2 + 14}$ for $x > 0$. Find the limit at infinity $\lim_{x \to \infty} f(x)$.

A 143

B 126

C 150

D **140** ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Dividing numerator and denominator by x^2 gives 140. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 140. **[A1]**

11 MCQ

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2.
During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm
.

A 50π

B $\frac{36\pi}{5}$

C 20π ✓

D 11π

WORKED SOLUTION & VERIFICATION

1. **Express the surface area constraint:**
The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome:

$$S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$$

Given $S = 180\pi$:

$$3\pi r^2 + 2\pi rh = 180\pi \implies 3r^2 + 2rh = 180$$

2. **Find the value of r when $h = 16.5$ cm:**

$$3r^2 + 2(16.5)r = 180 \implies 3r^2 + 33r - 180 = 0$$

Dividing by 3:

$$r^2 + 11r - 60 = 0 \implies (r + 15)(r - 4) = 0$$

Since $r > 0$, we have $r = 4$ cm.

3. **Express the total volume in terms of r:**
The total volume is the sum of the cylinder's volume and the hemisphere's volume:

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

From the surface area equation, $2rh = 180 - 3r^2 \implies h = \frac{90}{r} - \frac{3}{2}r$. Substituting this into V:

$$V = \pi r^2 \left(\frac{90}{r} - \frac{3}{2}r\right) + \frac{2}{3}\pi r^3 = 90\pi r - \frac{3}{2}\pi r^3 + \frac{2}{3}\pi r^3 = 90\pi r - \frac{5}{6}\pi r^3$$

4. **Differentiate V with respect to r:**

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 90\pi - \frac{5}{2}\pi r^2$$

At $r = 4$:

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 90\pi - \frac{5}{2}\pi(16) = 90\pi - 40\pi = 50\pi$$

5. **Apply the chain rule for connected rates of change:**

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}r} \cdot \frac{\mathrm{d}r}{\mathrm{d}t} = 50\pi \times 0.4 = 20\pi \text{ cm}^3\text{s}^{-1}$$

Common Misconceptions:
- Choice B (50π) comes from finding $\frac{\mathrm{d}V}{\mathrm{d}r}$ but omitting multiplication by $\frac{\mathrm{d}r}{\mathrm{d}t}$.

- Choice C ($\frac{36\pi}{5}$) comes from forgetting to include the volume of the hemispherical top in V.
- Choice D (11π) arises from an algebraic factorization error yielding $r = 5$. [M1][A1]

12 MCQ

The variables x and y are connected by the equation $y = \sqrt{2x+5}$. Given that x increases at a constant rate of 6 units per second, find the rate of change of y with respect to time when $x = 2$.

A 1 unit per second

B $\frac{1}{3}$ units per second

C 2 units per second ✓

D 12 units per second

WORKED SOLUTION & VERIFICATION

We are given $y = (2x+5)^{\frac{1}{2}}$ and $\frac{dx}{dt} = 6$.

Step 1: Differentiate y with respect to x using the chain rule:

$$\frac{dy}{dx} = \frac{1}{2}(2x+5)^{-\frac{1}{2}} \cdot \frac{d}{dx}(2x+5) = \frac{1}{2\sqrt{2x+5}} \cdot 2 = \frac{1}{\sqrt{2x+5}}$$

Step 2: Evaluate $\frac{dy}{dx}$ at $x = 2$:

$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{1}{\sqrt{2(2)+5}} = \frac{1}{\sqrt{9}} = \frac{1}{3}$$

Step 3: Use the connected rates of change formula:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt} = \frac{1}{3} \times 6 = 2 \text{ units per second}$$

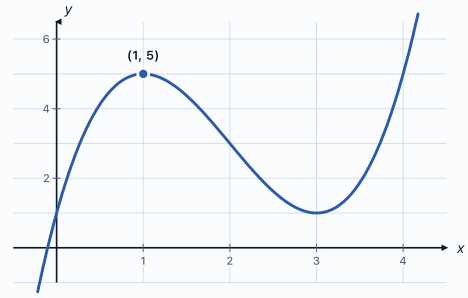
Misconception Analysis:

- 1 unit per second occurs if one forgets the derivative of the inner function $(2x+5)$, incorrectly taking $\frac{dy}{dx} = \frac{1}{2\sqrt{2x+5}} = \frac{1}{6}$.
- $\frac{1}{3}$ units per second is just $\frac{dy}{dx}$, omitting multiplication by $\frac{dx}{dt}$.
- 12 units per second arises from incorrectly differentiating $\sqrt{2x+5}$ as 2 and multiplying by 6. [M1][A1]

13 MCQ

The diagram shows part of the curve $y = ax^3 + bx^2 + 9x + 1$, where a and b are constants. The curve has a stationary point at $(1, 5)$.

Find the coordinates of the other stationary point on the curve.



A $(3, 0)$

B $(3, 1)$ ✓

C $(3, -1)$

D $(-3, 1)$

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** To find the coordinates of the other stationary point, we first determine the unknown parameters a and b.

Step 1: Use the given coordinates of the point on the curve

Since the point $(1, 5)$ lies on the curve:

$$5 = a(1)^3 + b(1)^2 + 9(1) + 1$$

$$5 = a + b + 10 \implies a + b = -5 \quad \text{— (Equation 1)}$$

Step 2: Differentiate to find $\frac{dy}{dx}$

$$\frac{dy}{dx} = 3ax^2 + 2bx + 9$$

Step 3: Apply the stationary condition at $x = 1$

At a stationary point, $\frac{dy}{dx} = 0$:

$$3a(1)^2 + 2b(1) + 9 = 0 \implies 3a + 2b = -9 \quad \text{— (Equation 2)}$$

Step 4: Solve the simultaneous equations

Multiplying Equation 1 by 2:

$$2a + 2b = -10$$

Subtracting this from Equation 2:

$$(3a + 2b) - (2a + 2b) = -9 - (-10) \implies a = 1$$

Substituting $a = 1$ into Equation 1:

$$1 + b = -5 \implies b = -6$$

Step 5: Find the other stationary point

Substitute $a = 1$ and $b = -6$ into the derivative:

$$\frac{dy}{dx} = 3x^2 - 12x + 9 = 0$$

Dividing by 3:

$$x^2 - 4x + 3 = 0 \implies (x-1)(x-3) = 0$$

Thus, the other stationary point occurs at $x = 3$.

Step 6: Calculate the corresponding y-coordinate

Substitute $x = 3$ back into the equation of the curve $y = x^3 - 6x^2 + 9x + 1$:

$$y = (3)^3 - 6(3)^2 + 9(3) + 1 = 27 - 54 + 27 + 1 = 1$$

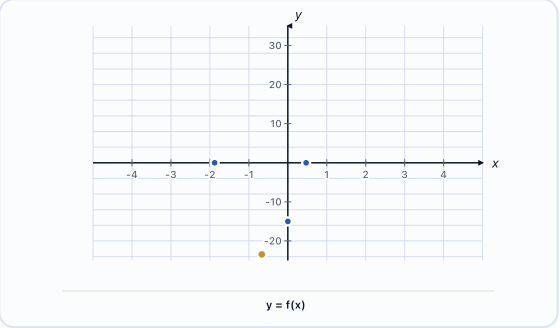
Therefore, the other stationary point is $(3, 1)$.

Common Trap: Substituting $x = 3$ into the derivative $\frac{dy}{dx}$ instead of the original curve equation yields $y = 0$, leading erroneously to $(3, 0)$. **[M1]**

- **Step 3 (Final Accuracy):** Verified exact result: $(3, 1)$. **[A1]**

14 MCQ

Let the differentiable function f be given by $f(x) = 17x^2 + 24x - 15$. Find the derivative $f'(15)$.



A 534 ✓

B 510

C 551

D 537

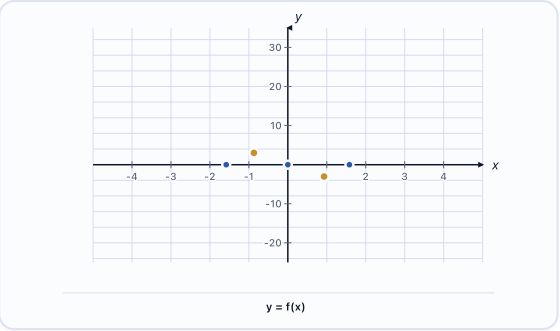
WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Differentiating gives $f'(x) = 34x + 24$, so $f'(15) = 534$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 534. **[A1]**

15 MCQ

The variables x and y are connected by the equation $y = 2x^3 - 5x$. Given that x increases at a constant rate of 4 units s^{-1}, find the rate of change of y at the instant when $x = 2$.



A 76 units s^{-1} ✓

B 24 units s^{-1}

C 19 units s^{-1}

D 28 units s^{-1}

WORKED SOLUTION & VERIFICATION

To find the rate of change of y with respect to time t, we apply the chain rule for connected rates of change:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

First, differentiate y with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$$

Substitute $x = 2$ into the derivative:

$$\left.\frac{dy}{dx}\right|_{x=2} = 6(2)^2 - 5 = 24 - 5 = 19$$

Given that $\frac{dx}{dt} = 4$ units s^{-1}, we calculate:

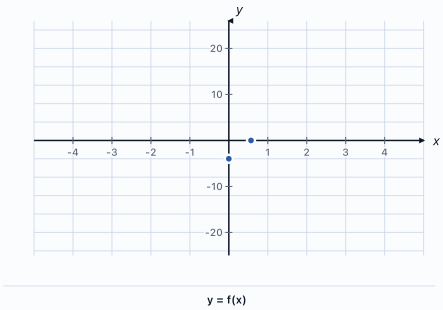
$$\frac{dy}{dt} = 19 \times 4 = 76 \text{ units } s^{-1}$$

Common errors:
- 19 units s^{-1} results from omitting multiplication by $\frac{dx}{dt}$.
- 28 units s^{-1} results from incorrectly differentiating $2x^3$ as $6x$, giving $(6(2) - 5) \times 4 = 28$.
- 24 units s^{-1} results from multiplying $y(2) = 6$ by $\frac{dx}{dt} = 4$. [M1][A1]

16 MCQ

The line $y = 7x - 4$ is tangent to the curve $y = ax^2 + \frac{b}{x}$ at the point where $x = 2$, where a and b are constants and $x \neq 0$.

Find the coordinates of the stationary point of the curve and determine its nature.



A $(1, 6)$, which is a local maximum

B $(1, 6)$, **which is a local minimum** ✓

C $(-1, -2)$, which is a local minimum

D $(2, 10)$, which is a local minimum

WORKED SOLUTION & VERIFICATION

1. **Find the point of tangency:**

Substitute $x = 2$ into the tangent line equation $y = 7x - 4$:

$$y = 7(2) - 4 = 10$$

Since the line touches the curve at $x = 2$, the point $(2, 10)$ lies on the curve $y = ax^2 + \frac{b}{x}$:

$$a(2)^2 + \frac{b}{2} = 10 \implies 4a + \frac{b}{2} = 10 \implies 8a + b = 20 \quad \text{— (1)}$$

2. **Use the gradient of the tangent:**

Differentiating $y = ax^2 + bx^{-1}$ gives:

$$\frac{dy}{dx} = 2ax - \frac{b}{x^2}$$

The gradient of the tangent line $y = 7x - 4$ is 7. At $x = 2$:

$$2a(2) - \frac{b}{2^2} = 7 \implies 4a - \frac{b}{4} = 7 \implies 16a - b = 28 \quad \text{— (2)}$$

3. **Solve for a and b:**

Adding equations (1) and (2):

$$(8a + b) + (16a - b) = 20 + 28 \implies 24a = 48 \implies a = 2$$

Substituting $a = 2$ into (1):

$$8(2) + b = 20 \implies b = 4$$

Thus, the equation of the curve is $y = 2x^2 + \frac{4}{x}$.

4. **Find the stationary point:**

Set $\frac{dy}{dx} = 0$:

$$4x - \frac{4}{x^2} = 0 \implies 4x^3 = 4 \implies x^3 = 1 \implies x = 1$$

Substitute $x = 1$ back into the curve equation:

$$y = 2(1)^2 + \frac{4}{1} = 6$$

So the stationary point is $(1, 6)$.

5. **Determine the nature:**

Find the second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(4x - 4x^{-2}) = 4 + 8x^{-3} = 4 + \frac{8}{x^3}$$

At $x = 1$:

$$\frac{d^2y}{dx^2} = 4 + \frac{8}{1^3} = 12 > 0$$

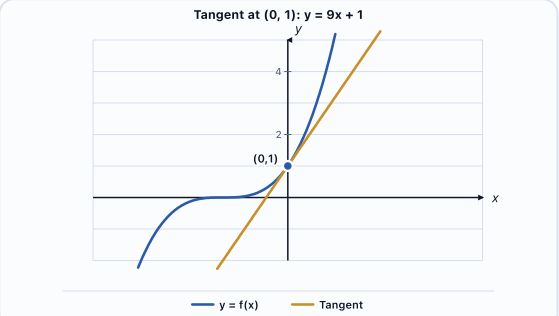
Since $\frac{d^2y}{dx^2} > 0$, the stationary point is a local minimum.

Common Traps:

- Choice B occurs when a student makes a sign error in the second derivative (e.g. $4 - 8x^{-3}$), resulting in $-4 < 0$ and erroneously concluding it is a maximum.
- Choice C occurs from a sign error when solving $4x - \frac{4}{x^2} = 0 \implies x^3 = -1 \implies x = -1$.
- Choice D mistakenly takes the given point of tangency $(2, 10)$ to be the stationary point. [M1][A1]

17 MCQ

Calculate the equation of the tangent line to the curve $y = (3x + 1)^3$ at the point where $x = 0$.



A $y = 9x + 1$ ✓

B $y = 9x + 3$

C $y = 9x - 1$

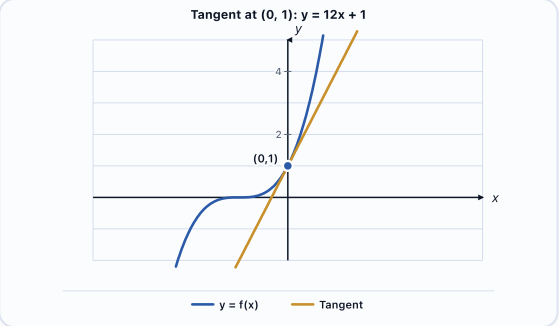
D $y = 3x + 1$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(3x + 1)^2 \cdot (3) = 9(3x + 1)^2$. At $x = 0$, slope $m = 9$. Equation: $y - 1 = 9(x - 0) \implies y = 9x + 1$. [M1][A1]

18 MCQ

Compute the equation of the tangent line to the curve $y = (4x + 1)^3$ at the point where $x = 0$.



A $y = 12x + 1$ ✓

B $y = 12x + 3$

C $y = 12x - 1$

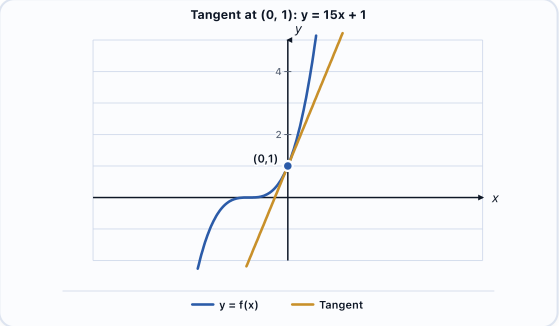
D $y = 4x + 1$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(4x + 1)^2 \cdot (4) = 12(4x + 1)^2$. At $x = 0$, slope $m = 12$. Equation: $y - 1 = 12(x - 0) \implies y = 12x + 1$. [M1][A1]

19 MCQ

Deduce the equation of the tangent line to the curve $y = (5x + 1)^3$ at the point where $x = 0$.



A $y = 5x + 1$

B $y = 15x + 3$

C $y = 15x + 1$ ✓

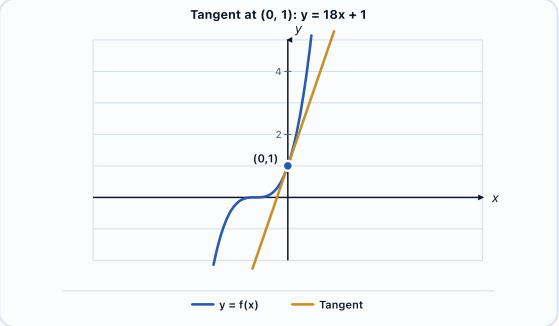
D $y = 15x - 1$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(5x + 1)^2 \cdot (5) = 15(5x + 1)^2$. At $x = 0$, slope $m = 15$. Equation: $y - 1 = 15(x - 0) \implies y = 15x + 1$. [M1][A1]

20 MCQ

Obtain the equation of the tangent line to the curve $y = (6x + 1)^3$ at the point where $x = 0$.



A $y = 6x + 1$

B $y = 18x - 1$

C $y = 18x + 3$

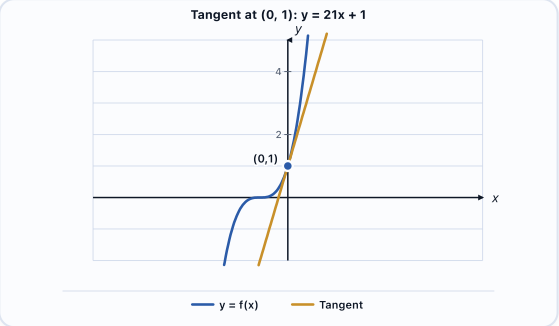
D $y = 18x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(6x + 1)^2 \cdot (6) = 18(6x + 1)^2$. At $x = 0$, slope $m = 18$. Equation: $y - 1 = 18(x - 0) \implies y = 18x + 1$. [M1][A1]

21 MCQ

Identify the equation of the tangent line to the curve $y = (7x + 1)^3$ at the point where $x = 0$.



A $y = 21x + 1$ ✓

B $y = 21x + 3$

C $y = 21x - 1$

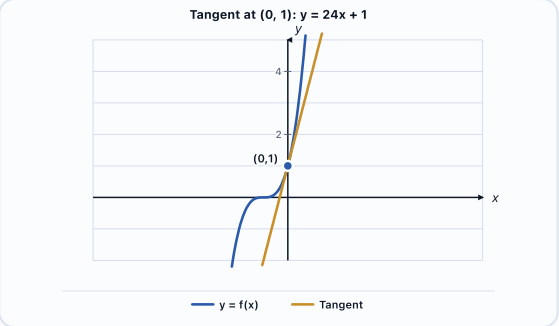
D $y = 7x + 1$

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(7x + 1)^2 \cdot (7) = 21(7x + 1)^2$. At $x = 0$, slope $m = 21$. Equation: $y - 1 = 21(x - 0) \implies y = 21x + 1$. [M1][A1]

22 MCQ

Establish the equation of the tangent line to the curve $y = (8x + 1)^3$ at the point where $x = 0$.



A $y = 24x - 1$

B $y = 8x + 1$

C $y = 24x + 3$

D $y = 24x + 1$ ✓

WORKED SOLUTION & VERIFICATION

At $x = 0$, $y = 1^3 = 1$. Derivative: $\frac{dy}{dx} = 3(8x + 1)^2 \cdot (8) = 24(8x + 1)^2$. At $x = 0$, slope $m = 24$. Equation: $y - 1 = 24(x - 0) \implies y = 24x + 1$. [M1][A1]

23 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 4$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	4

A 32π cm^3/s ✓

B $\frac{32}{2}\pi$ cm^3/s

C 64π cm^3/s

D 16π cm^3/s

WORKED SOLUTION & VERIFICATION

$V = \frac{4}{3}\pi r^3 \implies \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 4$, $\frac{dV}{dt} = 4\pi(4)^2(0.5) = 2\pi(16) = 32\pi$ cm^3/s. [M1][A1]

24 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 5$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	5

A 50π cm^3/s ✓

B 25π cm^3/s

C 100π cm^3/s

D $\frac{50}{2}\pi$ cm^3/s

WORKED SOLUTION & VERIFICATION

$V = \frac{4}{3}\pi r^3 \implies \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 5$, $\frac{dV}{dt} = 4\pi(5)^2(0.5) = 2\pi(25) = 50\pi$ cm^3/s. [M1][A1]

25 MCQ

Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 cm/s. What is the rate of change of volume $\frac{dV}{dt}$ when the radius is $r = 6$ cm?

Given Motion Parameters

Parameter 1	Parameter 2
0.5	6

A $\frac{72}{2}\pi$ cm^3/s

B 144π cm^3/s

C 36π cm^3/s

D 72π cm^3/s ✓

WORKED SOLUTION & VERIFICATION

$V = \frac{4}{3}\pi r^3 \implies \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$. At $r = 6$, $\frac{dV}{dt} = 4\pi(6)^2(0.5) = 2\pi(36) = 72\pi$ cm^3/s. [M1][A1]